Computer Programming Project 2025: 2D Ising Model

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Problem Definition

This project implements a Markov Chain Monte Carlo (MCMC) sampler for the 2D Ising model. The system consists of N^2 binary spins $s_i \in \{-1, +1\}$ arranged on a square lattice of side N with Periodic Boundary Conditions (PBC).

The cost function of the system is given by the Ising Hamiltonian:

$$\mathcal{H}(s) = -\frac{1}{2} \sum_i \sum_{j \in \mathcal{N}(i)} J_{ij} s_i s_j \tag{1}$$

where $\mathcal{N}(i)$ denotes the set of nearest neighbors of site i, and $J_{ij} = 1$ represents the ferromagnetic coupling constant.

We sample configurations from the Boltzmann distribution $P(s)$ at inverse temperature $\beta = 1/T$:

$$P(s) \propto e^{-\beta \mathcal{H}(s)} \tag{2}$$

The sampling is performed using the Metropolis algorithm with single-spin flip proposals ($s_i \to -s_i$). The energy variation associated with a flip is:

$$\Delta \mathcal{H} = 2 s_i \left(\sum_{j \in \mathcal{N}(i)} J_{ij} s_j \right) \tag{3}$$

A proposed move is accepted with probability:

$$P_{acc} = \min\{1, e^{-\beta \Delta \mathcal{H}}\} \tag{4}$$

A key observable monitored during the simulation is the magnetization per spin, defined as:

$$m(s) = \frac{1}{N^2} \sum_{i=1}^{N^2} s_i \tag{5}$$

This quantity helps characterize the phase transition between the ferromagnetic and paramagnetic states.

1 Task 1: Implementation of the Ising Class

1.1 Cost and Delta Cost Implementation

I implemented the `cost()` method to calculate the total energy of the system. This iterates over the entire lattice and sums the interactions, respecting PBC.

```
def cost(self):
    s = self.s
    N = self.N
    c = 0.
    for i, j in np.ndindex(N, N):
        # Sum interactions with 4 neighbors (up, down, left, right)
        c += s[i][j] * (s[i][(j+1)%N]+s[i][(j-1)%N]+
                        s[(i-1)%N][j]+s[(i+1)%N][j])
    return -0.5 * c
```

However, for the Metropolis algorithm, recomputing the full cost at every step is inefficient ($O(N^2)$). I implemented `compute_delta_cost()` to calculate only the energy difference ΔE caused by flipping a single spin ($O(1)$).

```
def compute_delta_cost(self, move):
    s = self.s
    N = self.N
    i, j = move
    # Calculate interaction with 4 neighbors using PBC
    neighbours = (s[i][(j+1)%N] + s[i][(j-1)%N] +
                  s[(i-1)%N][j] + s[(i+1)%N][j])

    # The energy change is +2 * s_i * neighbors
    return 2 * s[i][j] * neighbors
```

1.2 Cluster Detection Algorithm

I completed the 'find_clusters' function to analyze the geometric properties of the phases. The algorithm performs a Breadth-First Search (BFS) to identify connected components of spins with the same sign.

The logic I implemented iteratively expands the cluster by checking the four neighbours of the current spin. If a neighbor has the same sign and has not been visited yet, it is added to the `todo` set.

```
# Inside the while len(todo) > 0 loop:
s = todo.pop()
i, j = s // N, s % N # Decode 1D index to 2D

for di, dj in possible_configs:
    # Neighbour coordinates with Periodic Boundary Conditions
    ii = (i + di) % N
    jj = (j + dj) % N

    # Check if neighbor has the same spin
    if sample[ii, jj] == sign:
        ss = N * ii + jj # Encode to 1D index

        # Add to cluster if not already processed
        if mask[ss] and (ss not in done) and (ss not in todo):
            todo.add(ss)
            count += 1

done.add(s)
mask[s] = False
```

2 Task 2: Exploration of MCMC Parameters

2.1 Visualization of Regimes

To understand the behavior of the Ising model it is useful to visualise the lattice configurations for lattice sizes $N \in \{10, 25, 100\}$ and temperatures $T \in \{0.5, 2.5, 5.0\}$ using the `display()` method.

At low temperature ($T = 0.5$), the system exhibits the ferromagnetic phase. The spins are highly aligned, forming a single dominant domain. This confirms that for $T < T_C$, the system is able to find a uniform consensus.

At high temperature ($T = 5.0$), the system is in the paramagnetic phase. Thermal fluctuations overwhelm the magnetic interactions, resulting in random noise.

At the intermediate temperature ($T = 2.5$), which is close to the critical temperature ($T_C \approx 2.26$), we observe the formation of large clusters. Unlike the low-temperature case, the system has not settled into a single state, but domains of aligned spins of various sizes are visible.

2.2 Acceptance Rates and Burn-in

To ensure the Markov Chain samples from the equilibrium distribution, I investigated the evolution of the system's energy (cost) over time. Figure 2 displays the energy per spin for the largest lattice size, $N = 100$.

Dependence of Burn-in on T and N: The results clearly show that the minimal `burn_in` period is strongly dependent on temperature.

- At High T ($T = 5.0$), the system equilibrates almost immediately. The random initial configuration is already close to the disordered equilibrium state.
- At Low T ($T = 0.5$), the energy decreases rapidly but fails to plateau within the observed window. This steep downward slope indicates the system is still navigating towards the ferromagnetic ground state. This confirms that for low temperatures (and large N), the burn-in time must be orders of magnitude larger to avoid sampling from non-equilibrium states.

Acceptance Rates: I analyzed the Metropolis acceptance rate as a function of temperature for varying lattice sizes ($N \in \{10, 25, 100\}$).

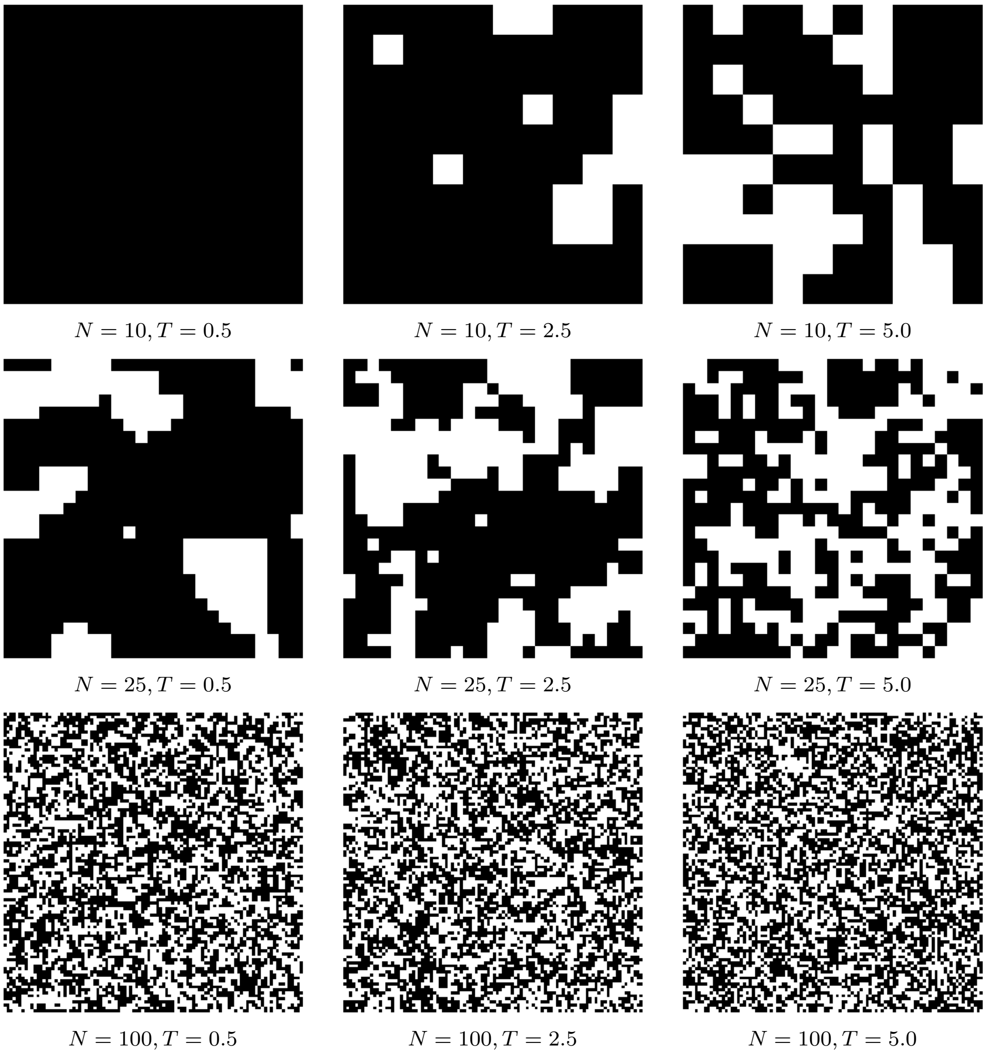

Figure 1: Snapshots of the 2D Ising model. Columns: $T = 0.5, 2.5, 5.0$.s Rows: $N = 10, 25, 100$.

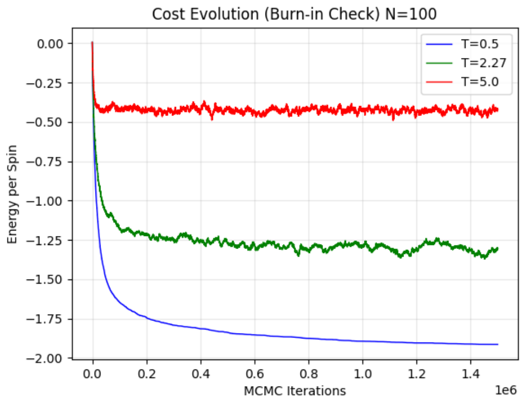

Figure 2: Energy relaxation for $N = 100$. While high temperatures (red) equilibrate instantly, the low-temperature system (blue) is still relaxing after 2000 iterations, indicating a need for a much longer burn-in.

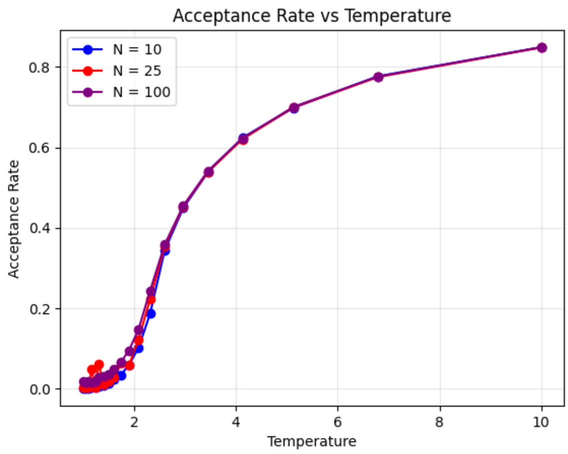

Figure 3: Acceptance Rate vs Temperature. The perfect overlap of the curves confirms that the acceptance rate is independent of the system size N.

As shown in Figure 3, the acceptance rate increases with temperature.

- At low T, thermal fluctuations are suppressed, and most moves increasing energy are rejected ($P_{acc} \to 0$).
- At high T, thermal energy dominates energy barriers, and the acceptance rate approaches unity.
- Independence from N: The curves for $N = 10$, $N = 25$, and $N = 100$ overlap perfectly. This demonstrates that the acceptance rate is an *intensive* quantity; it depends only on the local spin interactions and temperature, not on the macroscopic size of the lattice.

Behavior near T_C: Near the critical temperature ($T \approx 2.26$), we observe intermediate acceptance rates. However, efficiency is limited here because, as noted in the guidelines, samples become strongly correlated near the critical temperature. Even with reasonable acceptance rates, the system evolves slowly, requiring a significantly larger wait parameter to generate statistically independent samples.

3 Task 3: Statistical Properties

3.1 Magnetization and Critical Exponents

I investigated the magnetic phase transition by measuring the average magnetization $\langle|m|\rangle$ as a function of temperature for a lattice size of $N = 50$. The results are shown in Figure 4.

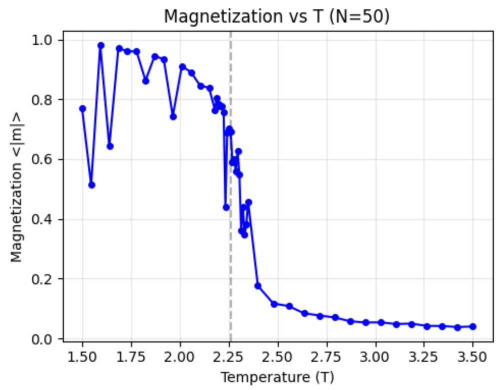

Figure 4: Magnetization vs Temperature for $N = 50$. The system exhibits a spontaneous magnetization close to 1.0 at low temperatures, which rapidly collapses to zero as the temperature approaches the critical point $T_C \approx 2.26$.

Analysis of the Phase Transition

- Low T ($T < 2.0$): The system is in the ferromagnetic phase. The strong interaction between spins maintains a near-perfect alignment, resulting in $\langle|m|\rangle \approx 1$.
- Critical Region ($2.0 < T < 2.3$): Thermal fluctuations begin to overcome the magnetic ordering. The magnetization drops sharply, indicating the start of the phase transition.
- High T ($T > 2.3$): The system enters the paramagnetic phase where thermal noise dominates, and the net magnetization vanishes.

Estimation of β

To quantify the "sharpness" of this transition, we analyzed the critical exponent β, which governs the power-law decay of magnetization near the critical point: $m \propto (T_C - T)^{\beta}$. We performed a linear regression on the log-log plot of the magnetization data, as shown in the left panel of Figure 5.

The analysis yields an experimental value of:

$$\beta_{exp} \approx 0.199 \tag{6}$$

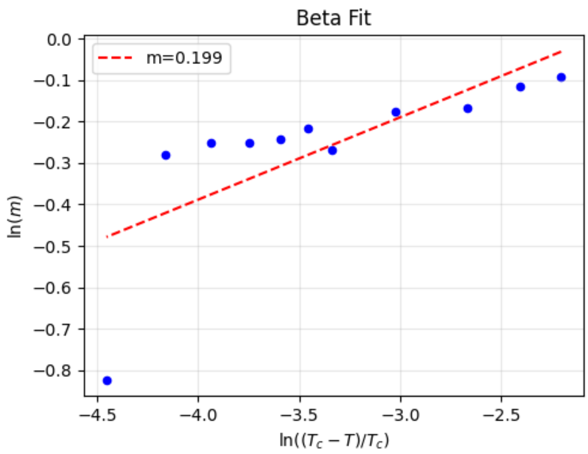

Figure 5: Estimation of the critical exponent β for a lattice size of $N = 50$.

Comparison with Magnetization Data: The obtained value of $\beta \approx 0.199$ is consistent with the visual profile of the magnetization curve in Figure 4.

- A smaller β (like the theoretical value $\beta_{theo} = 0.125$) corresponds to a curve that stays high for longer and drops very vertically near T_C.
- Our experimental value ($\beta \approx 0.199$) suggests that the magnetization drops off a bit more gradually than the sharp theoretical prediction. This softening of the curve is a side effect of simulating a finite grid ($N = 50$). Because our lattice is not infinite, the clusters of spins can only grow so large before hitting the boundaries. This limitation prevents the phase transition from being perfectly sharp, resulting in a slightly higher exponent than expected for an infinite system.

3.2 Susceptibility and Critical Exponent γ (Task 3.2)

We examined the magnetic susceptibility χ as a function of temperature. This quantity measures how easily the system can be magnetized by an external field. The results for $N = 50$ are presented in Figure 6.

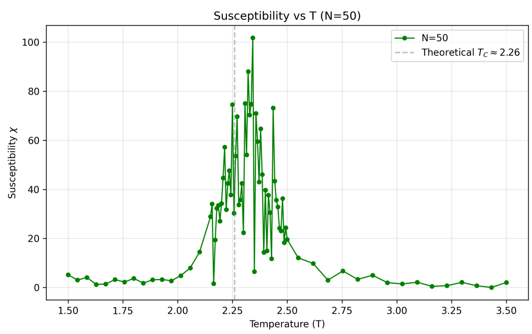

Figure 6: Susceptibility χ vs Temperature for $N = 50$. The value spikes dramatically near the critical temperature $T_C \approx 2.26$.

Behavior of Susceptibility

The graph shows three distinct regions:

- Stable Regions (Low and High T): Far from the critical temperature, the susceptibility is low. The system is stable and does not fluctuate much.
- The Peak (Critical Region): As the temperature approaches T_C, the susceptibility shoots up. This indicates that the spins are becoming extremely sensitive and unstable.
- Noise in the Peak: The peak in Figure 6 appears "noisy" with several jagged spikes. This is consistent with sampling expectations, near T_C, consecutive samples in the simulation become *strongly correlated*. Because the system evolves so slowly in this region, it is computationally difficult to get perfectly independent data points, leading to the large fluctuations observed in the graph.

Estimation of γ

Theory predicts that susceptibility grows as a power law: $\chi \propto |T - T_C|^{-\gamma}$. I estimated the exponent γ by plotting the natural logarithm of χ against the logarithm of the reduced temperature.

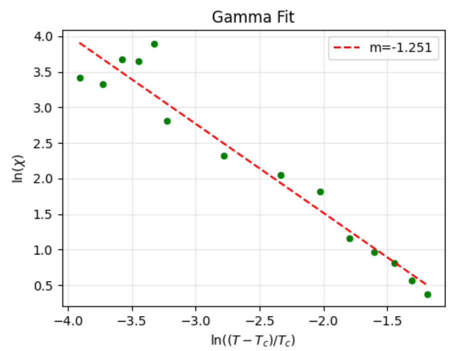

Figure 7: Log-Log fit for the susceptibility exponent γ for N = 50.

As shown in Figure 7, the linear fit gives a slope of approximately -1.25. Since the relationship is $\ln(\chi) \approx -\gamma \ln(\frac{T-T_c}{T_c})$, this gives an experimental value of:

$$\gamma_{exp} \approx 1.25 \tag{7}$$

Comparison with Theory: The theoretical value for an infinite 2D Ising model is $\gamma_{theo} = 1.75$. Our lower value (≈ 1.25) can be expected for a finite lattice ($N = 50$). In a real infinite system, the susceptibility would diverge to infinity. In our small grid, the clusters of spins hit the boundaries of the box, which limits how large the susceptibility can grow. This rounding off of the peak results in a smaller measured exponent.

3.3 Cluster Size Distributions

To investigate the microscopic structure of the system, I analyzed the distribution of cluster sizes. A "cluster" is defined as a group of neighboring spins with the same alignment. I collected data for three different lattice sizes ($N \in \{10, 50, 100\}$) at three distinct temperatures: Low ($T = 0.5$), Critical ($T \approx 2.26$), and High ($T = 5.0$).

The results are plotted in Figure 8 using a log-log scale. I define $P(s)$ as the probability density, normalised such that the sum of probabilities over all cluster sizes equals 1. This normalization results in small decimal values on the y-axis but allows me to fairly compare systems with different total numbers of spins.

Analysis by Temperature Regime

1. Low Temperature ($T = 0.5$, Top Row): In the ferromagnetic phase, the thermal energy is too weak to flip spins against their neighbors.

- I observe a single isolated bar at the far right of the x-axis.
- This indicates that almost the entire lattice forms one big, system-spanning cluster. The size of this cluster is $s \approx N^2$ (e.g., for $N = 50$, the peak is near 2500).

2. High Temperature ($T = 5.0$, Bottom Row): In the paramagnetic phase, thermal fluctuations dominate, randomizing the spins.

- The distribution is heavily concentrated on the far left (small s).
- The probability drops off extremely rapidly (exponentially). This means that large clusters are exponentially unlikely to form; the system is made up almost entirely of tiny disconnected groups of spins.

3. Critical Temperature ($T \approx 2.26$, Middle Row): At the critical point, the system is in a unique state of transition.

- Mathematically, a straight line on a log-log scale implies a **Power Law** distribution: $P(s) \propto s^{-\tau}$.
- This behavior indicates **scale invariance**: clusters of all sizes, from tiny pairs to large groups—coexist simultaneously. Unlike the high-T or low-T phases, there is no typical cluster size at the critical point.

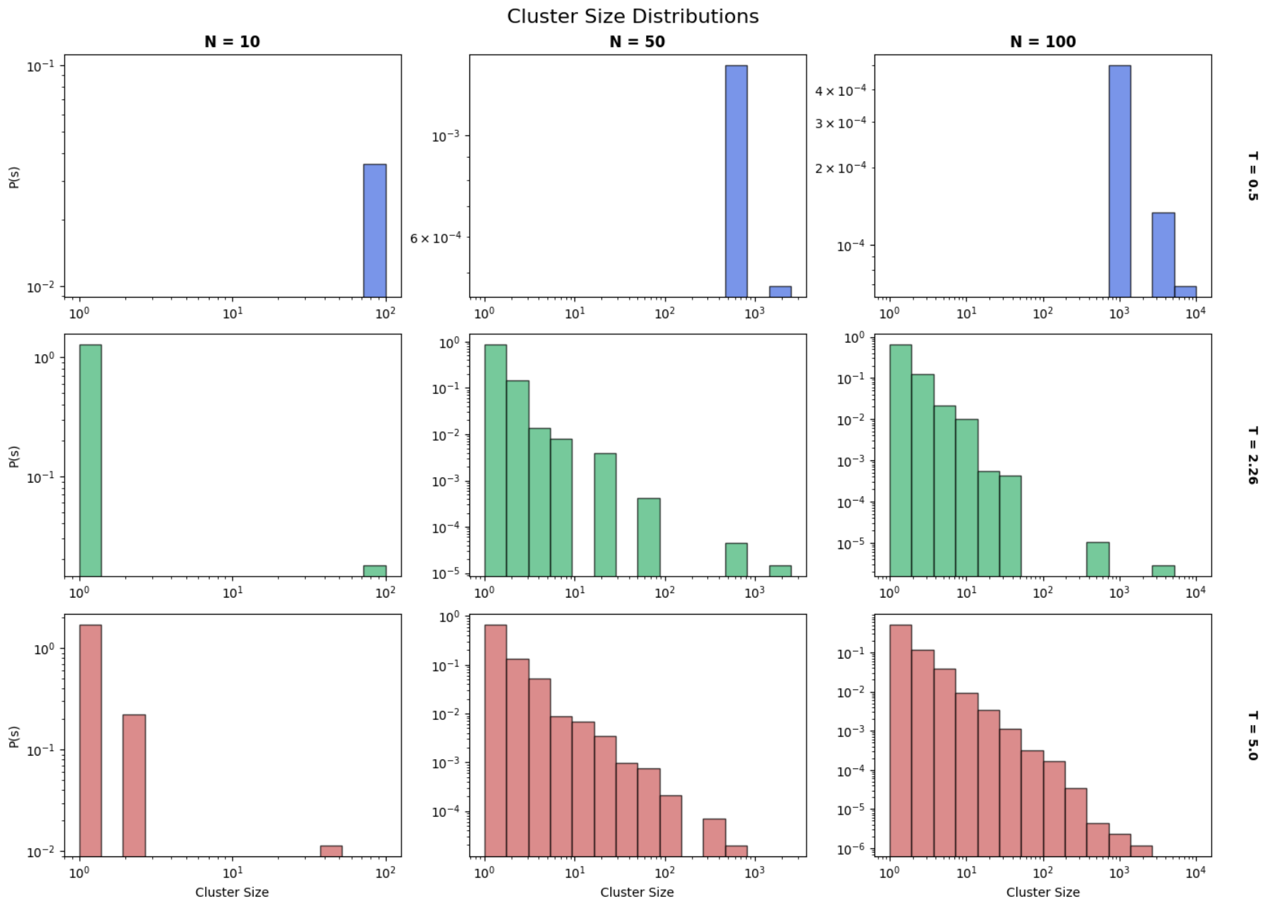

Figure 8: Probability density $P(s)$ of cluster sizes s. Top Row ($T = 0.5$): Single giant cluster dominates. Middle Row ($T \approx 2.26$): Power-law scaling (straight line). Bottom Row ($T = 5.0$): Exponential decay (tiny clusters).